Cross-disease reanalysis of IDI2-AS1 and IL5 in type-2 allergic disease cohorts

Five public bulk RNA-seq cohorts, 856 samples, one analysis framework

STAGE 1 — Datasets

- Asthma — GSE152004 — n = 695 — Nasal epithelium
- Atopic dermatitis — GSE121212 — n = 65 — Skin biopsy
- EoE — GSE246323 — n = 10 — Esophagus
- EoE — GSE58640 — n = 16 — Esophagus
- CRSwNP — GSE136825 — n = 70 — Polyp / control turbinate

STAGE 2 — Analysis

DESeq2 + apeglm (raw counts) / limma-trend (FPKM) → Random-effects meta-analysis (REML, unshrunken estimates) → Sample-level Spearman IDI2-AS1 ↔ IL5 → Signature-adjusted partial correlation (eosinophil, Th2, neutrophil) → Effect decomposition (asthma cohort, 10,000 bootstraps)

STAGE 3 — Three in vivo questions

- Q1. Is IDI2-AS1 itself differentially expressed in disease tissue?
- Q2. Do IDI2-AS1 and IL5 covary at the sample level in the predicted (negative) direction?
- Q3. If the bulk signal departs from the in vitro prediction, can it be ascribed to cell composition?

STAGE 4 — Principal finding

No cohort showed significant IDI2-AS1 differential expression (pooled $\log_2$FC +0.02 [−0.21, 0.25]; 80 % power for a 1.2-fold change in asthma).
Sample-level IDI2-AS1 ↔ IL5 correlation is POSITIVE — opposite to the in vitro repressive direction.
In asthma, conditioning on eosinophil estimates attenuates it: the composition-associated component is significant under three overlapping expression-derived estimates (marker score, ssGSEA, xCell enrichment; all $p \le 0.0014$), and no residual association is detected, with or without case/control adjustment.

→ Bulk RNA-seq can neither support nor refute per-cell regulation of IL5 by IDI2-AS1.
→ Cell-resolved measurement and perturbation in IL-5-producing cells are the natural next step.